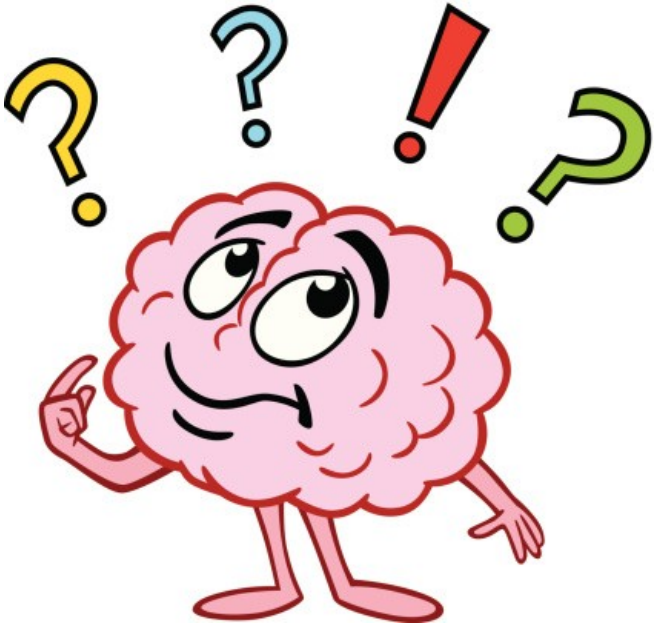
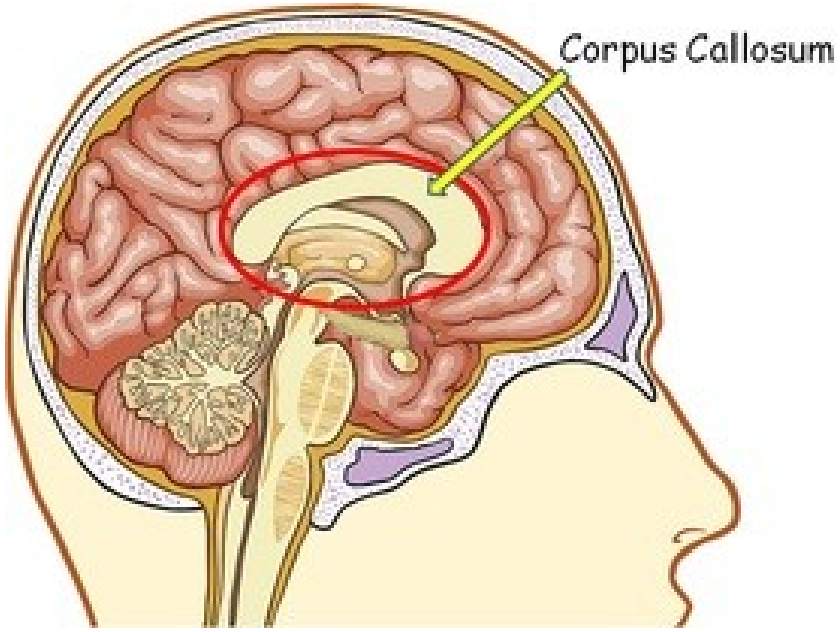
Central Nervous System – Brain



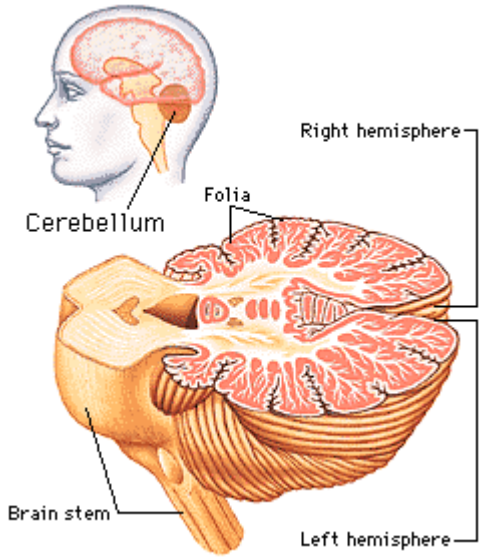
Corpus Callosum
Cerebellum
Hypothalamus
Medulla Oblongata
Spinal Cord

Corpus Callosum

- Wide band of nerve fibres
- At base of longitudinal fissure
- Allow 2 sides of cerebrum to communicate



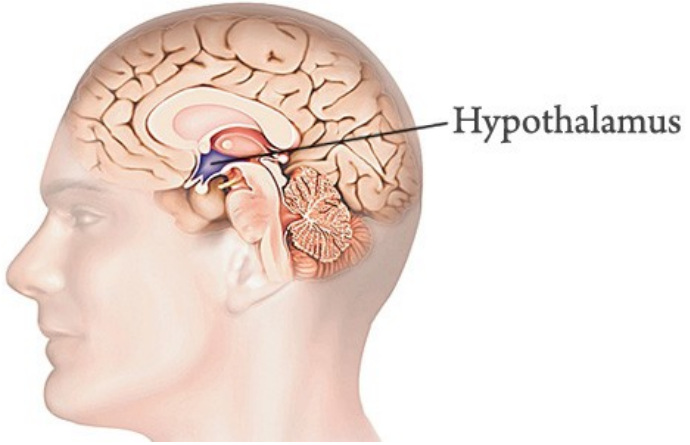
Cerebellum



- Under rear of cerebrum
- Highly folded surface
- Control over:
  - Balance
  - Posture
  - Fine coordination of voluntary muscle movement
- Receives input from inner ear, stretch receptors in skeletal muscles.
- “smooths” and coordinates movement.

Hypothalamus

- Deep in middle of brain
- Small but controls many activities
- Homeostatic control
  - Autonomic nervous system coordination
  - Body temperature
  - Food and water intake
  - Emotional responses
  - Coordination of parts of endocrine system
    - Metabolism
    - Growth
    - Reproduction
    - Responses to stress



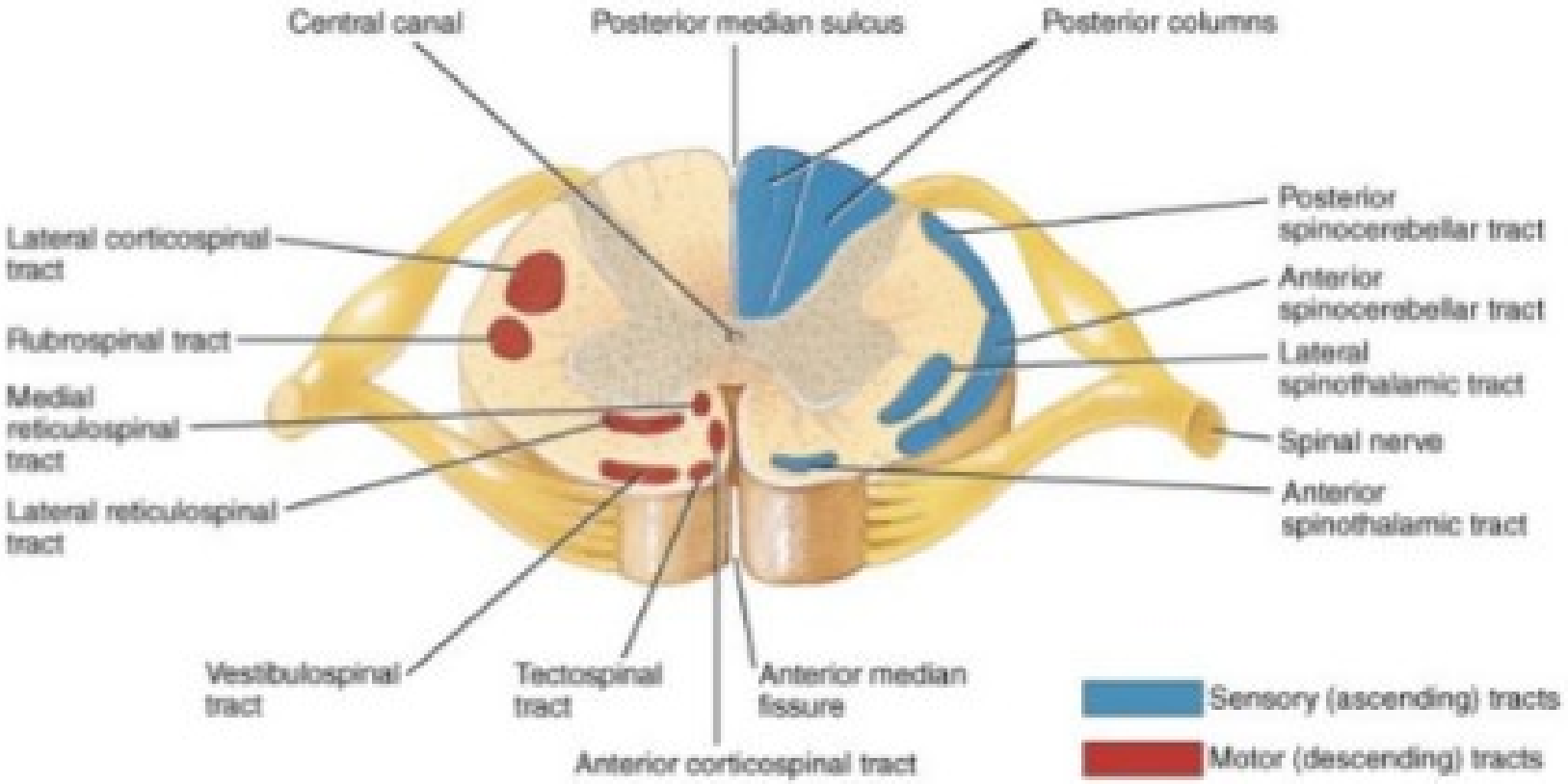
Medulla Oblongata

- Continuation of spinal cord at base of brain.
- Nerve fibres pass through on the way to other centres
- Also has role in coordinating key body functions:
  - **Cardiac centre:** rate and force of heartbeat
  - **Respiratory centre:** rate and depth of breathing
  - **Vasomotor centre:** diameter of blood vessels – blood pressure
  - Swallowing, coughing, sneezing, vomiting reflexes

- Interacts with hypothalamus and other higher brain centres to control homeostasis.

Spinal Cord

- Extends from base of skull down spinal column
- Heavily protected by bone, meninges, CSF
- Grey matter at centre, white matter at outside
- White matter in bundles
  - Ascending tracts – sensory axons
  - Descending tracts – motor axons
- Carries impulses up and down between brain and body
- Coordinates some reflexes



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Table 5.1 A summary of the functions of the main functional areas of the CNS

Structure	Function
Cerebral cortex	Higher-order functions such as thinking, reasoning, memory, learning, conscious awareness of surroundings
Corpus callosum	Communication between the two cerebral hemispheres
Cerebellum	Coordination of fine contractions of muscles resulting in smooth movements and the maintenance of posture and balance
Hypothalamus	Homeostasis; regulation of the heart, digestive system, appetite, thirst, metabolism, body temperature, response to fear or anger
Medulla oblongata	Under the influence of the hypothalamus, regulates the heart, breathing and diameter of blood vessels
Spinal cord	Provides a pathway for communication between muscles and glands and the brain. Integration of automatic, protective reflexes